

Ordinary Least Squares and the Normal Equations — Complete

These notes develop the normal equations for ordinary least squares (OLS). The goal is to connect derivative-based optimization from calculus with the matrix notation used in regression.

all completed responses and added explanations are shown in red.

Learning goals

By the end of these notes, you should be able to write the residual sum of squares in matrix form, apply the matrix-derivative rules needed for OLS, derive the normal equations, verify that their solution minimizes the loss, and interpret them geometrically.

1. Setup and notation

We observe n responses and use p coefficients, including an intercept coefficient when the model has an intercept.

Symbol	Dimensions	Meaning
X	$n \times p$	Design matrix; row i is $x_i^\top$
y	$n \times 1$	Response vector
β	$p \times 1$	Candidate coefficient vector
$X\beta$	$n \times 1$	Fitted-value vector
$r = y - X\beta$	$n \times 1$	Residual vector

The linear model is $y = X\beta + \varepsilon$. For a candidate β ,

$$\hat{y} = X\beta, \quad r = y - X\beta.$$

If the model contains an intercept, the first column of X is $\mathbf{1}$.

Quick dimension check

Expression	Dimensions
$X\beta$	$n \times 1$
$X^\top y$	$p \times 1$
$X^\top X$	$p \times p$
$\beta^\top X^\top y$	1×1 (a scalar)
$XX^\top$	$n \times n$

2. From scalar to matrix least squares

In scalar notation, the residual sum of squares is

$$\text{RSS}(\beta) = \sum_{i=1}^n (y_i - x_i^\top \beta)^2.$$

The same objective in matrix notation is

$$L(\beta) = \|y - X\beta\|_2^2 = (y - X\beta)^\top (y - X\beta).$$

OLS chooses $\hat{\beta}$ to minimize $L(\beta)$.

What dimensions must $\nabla_\beta L(\beta)$ have? Why?

It must be $p \times 1$, the same shape as β , because its entries give the rate of change of the scalar loss with respect to each coefficient.

Think first

1. Why is $L(\beta)$ always nonnegative?

It is a sum of squared residuals, equivalently a squared Euclidean norm.

2. Why does minimizing $L(\beta)/n$ give the same $\hat{\beta}$?

Dividing by the positive constant n does not change which value of β minimizes the objective.

3. Does defining this problem require normally distributed errors?

No. Normality is not needed to define or minimize the least-squares loss.

3. Matrix-derivative toolbox

Let $\mathbf{b} \in \mathbb{R}^p$ be the column vector with respect to which we differentiate. In this toolbox, **bold lowercase letters denote vectors**, ordinary lowercase letters denote scalar constants, and uppercase letters denote constant matrices. Thus $\mathbf{a}$ and $\mathbf{b}$ are vectors; c is a scalar constant; and A is a constant matrix. We use the **column-gradient convention**, so $\nabla_{\mathbf{b}} f$ is $p \times 1$ when f is scalar-valued. Each rule assumes compatible dimensions; take $A \in \mathbb{R}^{p \times p}$ and $\mathbf{a} \in \mathbb{R}^p$.

Function $f(\mathbf{b})$	Gradient $\nabla_{\mathbf{b}} f(\mathbf{b})$	Note
c	0	Constant rule
$\mathbf{a}^\top \mathbf{b}$	$\mathbf{a}$	Linear form
$\mathbf{b}^\top \mathbf{a}$	$\mathbf{a}$	Same scalar as $\mathbf{a}^\top \mathbf{b}$
$\mathbf{b}^\top A \mathbf{b}$	$(A + A^\top) \mathbf{b}$	General quadratic form
$\mathbf{b}^\top A \mathbf{b}$, if $A = A^\top$	$2A \mathbf{b}$	Symmetric quadratic form

4. Deriving the normal equations

Step 1: Expand the loss

Recall that $(AB)^\top = B^\top A^\top$.

Expanding the product gives

$$\begin{aligned} L(\beta) &= (\mathbf{y} - X\beta)^\top (\mathbf{y} - X\beta) \\ &= \mathbf{y}^\top \mathbf{y} - \mathbf{y}^\top X\beta - \beta^\top X^\top \mathbf{y} + \beta^\top X^\top X\beta. \end{aligned}$$

Every final term is a 1×1 scalar.

Step 2: Recognize equal scalar terms

$$\mathbf{y}^\top X\beta \text{ has dimensions } 1 \times 1 \implies \mathbf{y}^\top X\beta = (\mathbf{y}^\top X\beta)^\top = \beta^\top X^\top \mathbf{y}.$$

Thus the expanded loss can be written as

$$L(\beta) = y^\top y - 2\beta^\top X^\top y + \beta^\top X^\top X \beta.$$

Step 3: Differentiate term by term

$$\begin{aligned}\nabla_\beta L(\beta) &= \nabla_\beta(y^\top y) - 2\nabla_\beta(\beta^\top X^\top y) + \nabla_\beta(\beta^\top X^\top X \beta) \\ &= \mathbf{0} - 2X^\top y + 2X^\top X \beta \\ &= -2X^\top (y - X\beta).\end{aligned}$$

The gradient is $p \times 1$, as required.

Step 4: Apply the first-order condition

At an unconstrained differentiable minimum, the gradient is zero:

$$-2X^\top (y - X\hat{\beta}) = 0.$$

Dividing by -2 and rearranging gives

$$\boxed{X^\top X \hat{\beta} = X^\top y} \quad (\text{normal equations}).$$

These are called **normal equations** because they state that the residual is normal (perpendicular) to the columns of the design matrix—not because they require normally distributed errors.

Step 5: Solve for $\hat{\beta}$

If X has full column rank, then $X^\top X$ is invertible:

$$\begin{aligned}(X^\top X)^{-1} X^\top X \hat{\beta} &= (X^\top X)^{-1} X^\top y \\ \boxed{\hat{\beta} &= (X^\top X)^{-1} X^\top y}.\end{aligned}$$

Full column rank is needed only at this inversion step. The normal equations themselves can still be obtained when X is rank deficient.

5. Why the critical point is a minimum

A zero gradient identifies a critical point, but not whether it is a minimum, maximum, or saddle point.

$$H(\beta) = \nabla_\beta^2 L(\beta) = 2X^\top X.$$

For any nonzero $v \in \mathbb{R}^p$,

$$v^\top X^\top X v = (Xv)^\top (Xv) = \|Xv\|_2^2.$$

This quantity is always nonnegative. If X has full column rank and $v \neq 0$, then $Xv \neq 0$, so the quantity is strictly positive. Thus $X^\top X$ is positive definite, $L(\beta)$ is strictly convex, and $\hat{\beta}$ is the unique global minimizer.

If X is rank deficient, a least-squares minimum still exists, but the coefficient vector need not be unique. Different coefficient vectors may produce the same fitted values.

6. Derivation assumptions versus inference assumptions

Define/derive OLS	Obtain a unique inverse-form solution	Conduct statistical inference
Fitted values have the form $X\beta$	X has full column rank	$\mathbb{E}[\varepsilon \mid X] = 0$
Entries of X and y are finite	Equivalently, $X^\top X$ is invertible	$\text{Var}(\varepsilon \mid X) = \sigma^2 I_n$
Full rank is not required to state the normal equations		Normal errors are additionally used for exact finite-sample inference

Does deriving the normal equations require normal errors? Explain.

No. Normal errors are not needed to define the least-squares criterion, differentiate it, obtain the normal equations, or calculate an OLS solution. They are relevant only to particular inferential results.

7. Small example: completed calculation

Suppose

$$X = \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \end{bmatrix}, \quad y = \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}.$$

Part A: Build the normal equations

$$X^\top X = \begin{bmatrix} 3 & 3 \\ 3 & 5 \end{bmatrix}, \quad X^\top y = \begin{bmatrix} 5 \\ 6 \end{bmatrix}.$$

The resulting system is

$$\begin{cases} 3\beta_0 + 3\beta_1 = 5, \\ 3\beta_0 + 5\beta_1 = 6. \end{cases}$$

Part B: Solve and check

Subtracting the first equation from the second gives $2\beta_1 = 1$, so $\hat{\beta}_1 = \frac{1}{2}$. Substituting back gives $\hat{\beta}_0 = \frac{7}{6}$.

$$\hat{\beta} = \begin{bmatrix} \frac{7}{6} \\ \frac{1}{2} \end{bmatrix}, \quad \hat{y} = \begin{bmatrix} \frac{7}{6} \\ \frac{5}{3} \\ \frac{13}{6} \end{bmatrix}, \quad \hat{r} = \begin{bmatrix} -\frac{1}{6} \\ \frac{1}{3} \\ -\frac{1}{6} \end{bmatrix}.$$

$$X^\top \hat{r} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}, \quad \mathbf{1}^\top \hat{r} = 0.$$

8. Wrap-up and computation

The formula involving $(X^\top X)^{-1}$ is valuable for understanding the derivation. In numerical work, software generally uses a QR decomposition or singular value decomposition (SVD) instead of explicitly forming the inverse. These approaches are more stable when predictors are highly correlated or have very different scales.